

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020**(CBCSS—UG)****Mathematics****MEC 2C 02—MATHEMATICAL ECONOMICS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MEC 2C 02—MATHEMATICAL ECONOMICS
(Multiple Choice Questions for SDE Candidates)

1. The law which studies the direct relationship between price and quantity supplied of a commodity is :
 - (A) Law of demand.
 - (B) Law of variable proportion.
 - (C) Law of supply.
 - (D) None of the above.
2. When price rises, quantity supplied ?
 - (A) Expands.
 - (B) Falls.
 - (C) Increases.
 - (D) Unchanged.
3. When a percentage in price results in equal change in quantity supplied, it is called :
 - (A) Elastic supply.
 - (B) Perfectly inelastic.
 - (C) Elasticity of supply.
 - (D) Unitary elastic supply.
4. When supply of a commodity decreases on a fall in its price, it is called :
 - (A) Expansion of supply.
 - (B) Increase in supply.
 - (C) Contraction of supply.
 - (D) Decrease in supply.
5. At what point does total utility starts diminishing ?
 - (A) When marginal utility is positive.
 - (B) When it remains constant.
 - (C) When marginal utility is increasing.
 - (D) When marginal utility is negative.
6. Market which have two firms are known as :
 - (A) Oligopoly.
 - (B) Monopoly.
 - (C) Duopoly.
 - (D) Perfect competition.
7. In perfect competition a firm increases profit when _____ exceeds _____.
 - (A) TC, TR.
 - (B) MC, MR.
 - (C) AR, AC.
 - (D) TR, TFC.

8. If the coefficient of income elasticity of demand is higher than 1 and the revenue increases, the share of expenditures for commodity X in total expenditure : .
- (A) Will increase. (B) Will decrease.
(C) Will remain constant. (D) Can not be determined.
9. Suppose the price of a good decreases by 10 % and the quantity demanded for a certain period of time increases by 15 %. In these conditions :
- (A) The revenues earned by producers decrease.
(B) The revenues earned by producers increase.
(C) The revenues are not influenced in any way.
(D) The company's expenses rise.
10. Which of the following statements is false ?
- (A) Perfect competition involves many sellers of standardized products.
(B) Monopolistic competition involves many sellers of homogeneous products.
(C) The oligopoly involves several producers of standardized or differentiated products.
(D) Monopoly involves a single product for which there are no close substitutes.
11. A basic solution in a LPP is a _____ if it is feasible.
- (A) Basic feasible solution. (B) Non-basic feasible solution
(C) Both (A) and (B). (D) None.
12. A basic feasible solution is a basic solution whose variables are _____.
- (A) Feasible. (B) Negative.
(C) Non-negative. (D) None.
13. The divergence between Lorenz curve and line of perfect equality can be measured by :
- (A) Gini coefficient. (B) Coefficient of variation.
(C) Both. (D) None.
14. In LPP, the simplex method was developed by :
- (A) Koopman. (B) G. B. Dantzig.
(C) Leontief. (D) None of these.

15. Any non-negative value of (x_1, x_2) is a feasible solution of the LPP if it satisfies all the _____.
(A) Non-negativity conditions. (B) Constraints.
(C) Objective function. (D) None.
16. LP is a quantitative technique of decision-making using _____ constraints.
(A) Inequality. (B) Equality.
(C) Both (A) and (B). (D) None.
17. In LPP we deal with _____ objectives.
(A) Many. (B) Two.
(C) Three. (D) Four.
18. One of the limitations of LPP is to satisfy the assumption of _____ of objective function and constraints.
(A) Certainty. (B) Continuity.
(C) Linearity. (D) None.
19. Every linear programming problem has a _____ associated with it.
(A) Dual problem. (B) Assignment problem.
(C) Both. (D) None.
20. Zero sum game is also referred to as :
(A) Constant sum game. (B) Negative sum game.
(C) Both. (D) None.

SECOND SEMESTER B.A./B.Sc. DEGREE EXAMINATION, APRIL 2020

(CBCSS—UG)

Mathematics

MEC 2C 02—MATHEMATICAL ECONOMICS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer any number of questions.**Maximum marks : 20.*

1. Define Gini co-efficient.
2. What is meant by Price discrimination ?
3. What is a Non-negativity constraints ?
4. Explain the term Regression.
5. What is an Exogenous variable ?
6. Explain the Open input-output model.
7. State the Young's theorem.
8. What is meant by Local maxima and minima ?
9. Define a Symmetric matrix.
10. What is meant by Continuously differentiable functions ?
11. What is meant by Constrained optimization ?
12. Define Lorenz curve.

Section B

*Answer any number of questions.**Maximum marks : 30.*

13. From the data points, find the equation of the line which best fits the data points (1, 2), (3, 4), (5, 3), (6, 6).
14. Explain the sufficient and necessary conditions for unconstrained optimization.
15. Show whether the following function $x^4 + x^2 + 6xy + 3y^2$ has global minima or maxima.

Turn over

16. Examine whether the input-output system with the following co-efficient matrix is feasible :

$$\begin{bmatrix} 1/2 & 3/5 \\ 1/3 & 5/7 \end{bmatrix}.$$

17. Explain the optimization of a function with several equality constraints.
18. Compute the Hessian matrix of the function $4x^2y - 3xy^3 + 6x$.
19. Describe the Kuhn-Tucker formulation for a constrained minimization problem.

Section C

Answer any one (10 marks)

20. Explain the determination of equilibrium prices in an economy with two sectors using input-output model.
21. Explain the method of least squares and derive the normal equations.